arXiv:2412.06531v1 [cs.LG] 9 Dec 2024

UNRAVELING THE COMPLEXITY OF MEMORY IN RL AGENTS: AN APPROACH FOR CLASSIFICATION AND EVALUATION

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ABSTRACT

The incorporation of memory into agents is essential for numerous tasks within the domain of Reinforcement Learning (RL). In particular, memory is paramount for tasks that require the utilization of past information, adaptation to novel environments, and improved sample efficiency. However, the term "memory" encompasses a wide range of concepts, which, coupled with the lack of a unified methodology for validating an agent's memory, leads to erroneous judgments about agents' memory capabilities and prevents objective comparison with other memory-enhanced agents. This paper aims to streamline the concept of memory in RL by providing practical precise definitions of agent memory types, such as long-term versus short-term memory and declarative versus procedural memory, inspired by cognitive science. Using these definitions, we categorize different classes of agent memory, propose a robust experimental methodology for evaluating the memory capabilities of RL agents, and standardize evaluations. Furthermore, we empirically demonstrate the importance of adhering to the proposed methodology when evaluating different types of agent memory by conducting experiments with different RL agents and what its violation leads to.

1 INTRODUCTION

Reinforcement Learning (RL) effectively addresses various problems within the Markov Decision Process (MDP) framework, where agents make decisions based on immediately available information (Mnih et al., 2015; Badia et al., 2020). However, there are still challenges in applying RL to more complex tasks with partial observability.

To successfully address such challenges, it is essential that an agent is able to efficiently store and process the history of its interactions with the environment (Ni et al., 2021). Sequence processing methods originally developed for natural language processing (NLP) can be effectively applied to these tasks because the history of interactions with the environment can be represented as a sequence (Hausknecht & Stone, 2015; Esslinger et al., 2022; Samsami et al., 2024).

However, in many tasks, due to the complexity or noisiness of observations, the sparsity of events, the difficulty of designing the reward function, and the long duration of episodes, storing and retrieving important information becomes extremely challenging, and the need for memory mechanisms arises (Graves et al., 2016; Wayne et al., 2018; Goyal et al., 2022). Nevertheless, in the existing literature on RL, where the concept of "memory" is discussed, the definitions of memory are only defined in terms of the specific problem under consideration.

For example, in some works, memory is defined as the ability of an agent to effectively establish and use dependencies between events within a fixed-size sequence of tokens (context) in decision making (Esslinger et al., 2022; Ni et al., 2023; Grigsby et al., 2024). In other works, the term "memory" refers to the agent's ability to use out-of-context information through the use of various memory mechanisms (Parisotto et al., 2020; Lampinen et al., 2021; Cherepanov et al., 2024). In

the context of Meta-Reinforcement Learning (Meta-RL), however, the term "memory" is used to describe an agent's ability to use experience from other tasks or episodes to adapt to a new, previously unknown environment (Team et al., 2023; Kang et al., 2024a; Grigsby et al., 2024).

In this work, we treat memory as intrinsic attribute of memory-enhanced agents, linking the classification of memory types in RL directly to the characteristics of agent memory. These specific memory types can be assessed through experiments in memory-intensive environments. Our classification, based on temporal dependencies and the nature of the memorized information, provides a clear framework for distinguishing different memory types. This clear categorization is essential for fair comparisons between agents with similar memory mechanisms and for identifying limitations in an agent's memory architecture, aiding precise evaluations and improvements.

It is important to clarify that our goal is not to replicate the full spectrum of human memory. Instead, we draw from concepts of memory in neuroscience that are both widely recognized and intuitively applied within RL community, albeit without being explicitly defined or formalized (Fortunato et al., 2020; Ni et al., 2023; Kang et al., 2024b).

In summary, our contribution can be described as follows:

1. We formalize the definition of agent "memory" in RL: *long-term memory (LTM)* and *short-term memory (STM)*, *declarative memory* and *procedural memory* (section 5).
2. We introduce a decoupling of tasks that require an agent to have memory: *Memory Decision-Making* (*Memory DM*) and *Meta-Reinforcement Learning* (*Meta-RL*) (section 5).
3. We propose a generic experimental methodology for testing the LTM and STM capabilities of agents in Memory DM tasks (subsection 5.2).
4. We show that if the proposed experimental methodology is not followed, judgments about the agent's memory capabilities can become extremely incorrect (section 6).

2 Partially Observable Markov Decision Process

The Partially Observable Markov Decision Process (POMDP) is a generalization of the Markov Decision Process (MDP) that models sequential decision-making problems where the agent has incomplete information about the environment's state. POMDP can be represented as a tuple $\mathcal{M}_P = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, \mathcal{R}, \mathcal{Z} \rangle$, where $\mathcal{S}$ denotes the set of states, $\mathcal{A}$ is the set of actions, $\mathcal{O}$ is the set of observations and $\mathcal{Z} = \mathcal{P}(o_{t+1} \mid s_{t+1}, a_t)$ is an observation function such that $o_{t+1} \sim \mathcal{Z}(s_{t+1}, a_t)$. An agent takes an action $a_t \in \mathcal{A}$ based on the observed history $h_{0:t-1} = \{(o_i, a_i, r_i)\}_{i=0}^{t-1}$ and receives a reward $r_t = \mathcal{R}(s_t, a_t)$. It is important to note that state s_t is not available to the agent at time t. In the case of POMDPs, a policy is a function $\pi(a_t \mid o_t, h_{0:t-1})$ that uses the agent history $h_{0:t-1}$ to obtain the probability of the action a_t. Thus, in order to operate effectively in a POMDPs, an agent must have memory mechanisms to retrieve a history $h_{0:t-1}$. Partial observability arises in a variety of real-world situations, including robotic navigation and manipulation tasks, autonomous vehicle tasks, and complex decision-making problems.

3 Related Works

Researchers' interest in memory-enhanced RL agents is evident in the abundance of works proposing architectures with memory mechanisms and benchmarks (Osband et al., 2019; Morad et al., 2023; Pleines et al., 2023) for their validation (see Appendix C for details). However, despite the rather large number of works devoted to this topic, the term "memory" in RL still has multiple senses, and the selection of benchmarks and experiments is not always done correctly.

Thus, for instance, in Oh et al. (2016), memory is understood as the ability of an agent to store recent observations into an external buffer and then retrieve relevant information based on temporal context. In Lampinen et al. (2021), memory is the ability to store and recall desired information at long intervals. In Fortunato et al. (2020), memory refers to working and episodic memory (with short-term and long-term nature, respectively) from cognitive psychology and neuroscience, which allows an intelligent agent to use information from past events to make decisions in the present and future. Ni et al. (2023) describes two distinct forms of temporal reasoning: (working) memory and (temporal) credit assignment, where memory refers to the ability to recall a distant past event at the

current time. In Kang et al. (2024b) authors use the concept of reconstructive memory Bartlett & Kintsch (1995) discovered in psychology, which establishes a reflection process based on interaction.

4 Memory of Humans and Agents

Most works related to the concept of memory in RL use various principles from cognitive psychology and neuroscience such as long-term memory (Lampinen et al., 2021; Ni et al., 2023; Grigsby et al., 2024), working memory (Graves et al., 2014; Fortunato et al., 2020), episodic memory (Pritzel et al., 2017; Fortunato et al., 2020), associative memory (Parisotto & Salakhutdinov, 2017; Zhu et al., 2020), and others to introduce it. Despite the fundamental differences in these concepts, works on memory in RL often simplify these concepts to their inherent temporal scales (short-term memory and long-term memory). Regardless, the temporal scales are often presented qualitatively without clearly defining the boundaries between them. For example, many studies assume that remembering a few steps within an environment represents short-term memory, while remembering hundreds of steps represents long-term memory, without considering the relative nature of these concepts. This ambiguity between short-term and long-term memory can lead to a misattribution of an agent's memory capabilities and to an incorrect estimation of them when conducting experiments. To address this ambiguity, in this section we introduce formal definitions of agent memory in RL and its types, and propose an algorithm for designing an experiment to test agent memory in a correct way.

4.1 Memory in Cognitive Science

Human cognitive abilities that ensure adaptive survival depend largely on memory, which determines the accumulation, preservation, and reproduction of knowledge and skills (Parr et al., 2020; 2022). Memory exists in many forms, each of which relies on different neural mechanisms. Neuroscience and cognitive psychology distinguish memory by the temporal scales at which information is stored and accessed, and by the type of information that is stored. Abstracting from this distinction, a high-level definition of human memory is as follows: "***memory – is the ability to retain information and recall it at a later time***".

The definition aligns with the common understanding of memory in RL. Thus, we will use it to create terminology for various types of agent memory. In neuroscience, memory is categorized by temporal scale and behavioral manifestation. Typically, this leads to a distinction between *short-term memory*, which retains information for seconds, and *long-term memory*, which can last a lifetime (Davis & Squire, 1984). Additionally, memory is divided by behavioral manifestations into *declarative memory* (explicit) and *procedural memory* (implicit) (Graf & Schacter, 1985). Declarative memories can be consciously recalled, encompassing events and facts, while procedural memories are unconscious and relate to skills like skiing or driving.

In the following section, we introduce formal definitions of the above types of memory from neuroscience for RL tasks. Using these definitions, which are written in quantitative terms, we can uniquely classify the type of memory an agent has when using past information in decision making.

4.2 Memory in RL

The interpretation of memory in RL varies across studies. In some POMDPs, agents need to retain crucial information to make future decisions within a single environment. Here, memory typically encompasses two aspects: 1) the efficiency of establishing dependencies between events within a fixed time interval (e.g., transformer context (Esslinger et al., 2022; Ni et al., 2023)); and 2) the efficiency of establishing dependencies between events outside a fixed time interval (Parisotto et al., 2020; Sorokin et al., 2022).

Based on the neuroscience definitions outlined in subsection 4.1, the first interpretation aligns with short-term memory, while the second corresponds to long-term memory. Both interpretations are also closely related to declarative memory. In Meta-RL, memory typically refers to an agent's ability to leverage skills from different environments/episodes Team et al. (2023); Kang et al. (2024a), akin to procedural memory.

However, many studies fail to differentiate between agents with declarative and procedural memory, often treating Meta-RL tasks as a whole rather than focusing on decision-making based on past

information. For instance, when a paper asserts that an agent possesses long-term memory, it may only be tested on Meta-RL tasks based on MDPs. To clarify the concept of agent memory in RL, we provide formal definitions in this section.

In this paper, we primarily study an agent's memory, which is used to make current decisions based on past information within the same environment. Accordingly, our focus will be on declarative memory, specifically its short-term and long-term forms.

Memory and Credit Assignment. Papers exploring agent memory, particularly declarative memory, often distinguish between two concepts based on the temporal dependencies the agent must handle: *memory* and *credit assignment* (Osband et al., 2019; Mesnard et al., 2020; Ni et al., 2023). In Ni et al. (2023), the authors formally differentiate between two forms of temporal reasoning in RL: (working) memory and (temporal) credit assignment: "*memory* refers to the ability to recall a distant past event at the current time, while *credit assignment* refers to the ability to determine when the actions that merit current credit occurred" (Ni et al., 2023).

While distinct, these concepts both establish different temporal dependencies between related events. In this work, we focus on the agent's ability to form these dependencies, treating "memory" and "credit assignment" as a single entity. We will use the definition from subsection 4.1 to define memory generally. Notably, the definitions for "memory" also apply to "credit assignment", as they pertain solely to temporal dependencies rather than their essence.

5 Memory Decision Making

POMDP tasks that use agent memory can be divided into two main classes: *Meta Reinforcement Learning* (*Meta-RL*), which involves skill transfer across tasks, and *Memory Decision-Making* (*Memory DM*), which focuses on storing and retrieving information for future decisions.

This distinction is crucial: agents in Meta-RL use something like the procedural memory of subsection 4.1 to facilitate rapid learning and generalization, while those in Memory DM rely on something like declarative memory for current decision-making within the same environment. Despite these differences, many studies overlook behavioral manifestations and focus solely on temporal scales.

To introduce a definition for Memory DM tasks, we first need to introduce the definition of agent context length:

Definition 1. Agent context length ($K \in \mathbb{N}$) – is the maximum number of previous steps (triplets of (o, a, r)) that the agent can process at time t.

For example, an MLP-based agent processes one step at a time ($K = 1$), while a transformer-based agent can process a sequence of up to $K = K_{attn}$ triplets, where K_{attn} is determined by attention. Using the introduced Definition 1 for agent context length, we can introduce a formal definition for the Memory DM framework we focus on in this paper:

Definition 2. Memory Decision-Making (**Memory DM**) – is a class of POMDPs in which the agents decision-making process at time t is based on the history $h_{0:t-1} = \{(o_i, a_i, r_i)\}_{i=0}^{t-1}$ if $t > 0$ otherwise $h = \varnothing$. The objective is to determine an optimal policy $\pi^*(a_t \mid o_t, h_{0:t-1})$ that maps the current observation o_t and history $h_{0:t-1}$ of length t to an action a_t, maximizing the expected cumulative reward within a single POMDP environment $\mathcal{M}_P$: $J^\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]$, where T – episode duration, $\gamma \in [0, 1]$ – discount factor.

In the Memory DM framework (Definition 2), memory refers to the agent's ability to recall information from the past within a single environment and episode. In contrast, in the Meta-RL framework (see Appendix, Definition 7), memory involves recalling information about the agent's behavior from other environments or previous episodes. To distinguish these concepts, we adopt the definitions of *"Declarative memory"* and *"Procedural memory"* from subsection 4.1:

Definition 3 (**Declarative and Procedural memory in RL**)**.** Let n_{envs} be the number of training environments and n_{eps} the number of episodes per environment. Then,

1. **Declarative Memory** – a type of agent memory when an agent transfers its knowledge within a single environment and across a single episode within that environment:

$$\text{Declarative Memory} \iff n_{envs} \times n_{eps} = 1 \tag{1}$$

2. **Procedural Memory** – a type of agent memory when an agent transfers its skills across multiple environments or multiple episodes within a single environment:

$$\text{Procedural Memory} \iff n_{envs} \times n_{eps} > 1 \tag{2}$$

Here, "knowledge" refers to observable information like facts, locations, and events. In contrast, "skills" are pre-learned policies that an agent can apply across various tasks. Thus, the Memory DM framework validates the agent's declarative memory, while the Meta-RL framework validates its procedural memory (see Figure 1).

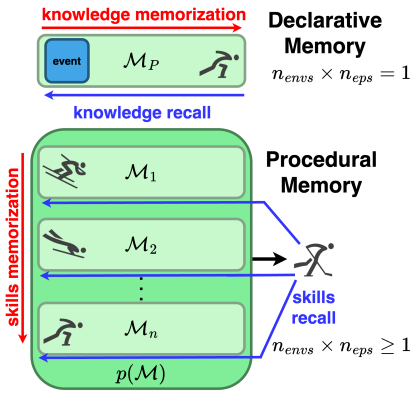


Figure 1: Declarative and procedural memory scheme. Red arrows show the information transfer for memorization, blue arrows show the direction of recall to the required information.

In subsection 4.2, we distinguished two classes of POMDPs: Memory DM, which requires declarative memory, and Meta-RL, which requires procedural memory. Within the Memory DM tasks, which are our primary focus, agent memory is categorized into long-term memory and short-term memory:

Definition 4 (**Memory DM types of memory**). Let K be the agent context length, $\alpha_{t_e}^{\Delta t} = \{o_i, a_i, r_i\}_{i=t_e}^{t_e+\Delta t}$ – an event of duration Δt that begins at $t = t_e$ and ends at $t = t_e + \Delta t$, and $\beta_{t_r}(\alpha_{t_e}^{\Delta t}) = a_t \mid (o_t, \alpha_{t_e}^{\Delta t})$ – a decision-making point (recall) at time $t = t_r$ based on the current observation o_t and information about the event $\alpha_{t_e}^{\Delta t}$. Let also $\xi = t_r - t_e - \Delta t + 1$ be the **correlation horizon**, i.e. the minimal time delay between the event $\alpha_{t_e}^{\Delta t}$ that supports the decision-making and the moment of recall of this event β_{t_r}. Then,

1. **Short-term memory (STM)** – an agent ability to utilize information about local correlations from the past within the agent context of length K at the time of decision making:

$$\text{Short-term memory} \iff \beta_{t_r}(\alpha_{t_e}^{\Delta t}) = a_t \mid (o_t, \alpha_{t_e}^{\Delta t}) \; \forall \, \xi = t_r - t_e - \Delta t + 1 \leq K$$

2. **Long-term memory (LTM)** – an agent ability to utilize information about global correlations from the past outside of the agent context of length K, during decision-making:

$$\text{Long-term memory} \iff \beta_{t_r}(\alpha_{t_e}^{\Delta t}) = a_t \mid (o_t, \alpha_{t_e}^{\Delta t}) \; \forall \, \xi = t_r - t_e - \Delta t + 1 > K$$

An illustration for the definitions of classifying Memory DM tasks into LTM and STM from Definition 4 is shown in Figure 2.

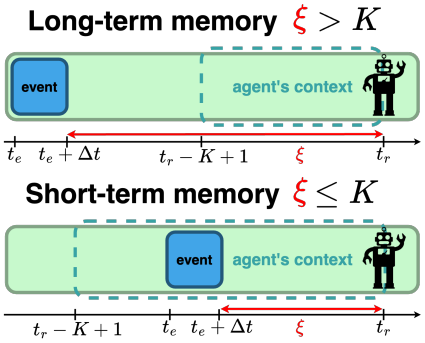


Figure 2: Long-term memory and short-term memory scheme. t_e – event used for decision-making start time, Δt – event duration, t_r – agent's recall time, K – agent's context length, ξ – correlation horizon. If an event is outside the context, long-term memory is needed for decision-making; if within the context, short-term memory suffices.

The two definitions of declarative memory encompass all work related to Memory DM tasks, where decisions are based on past information. Meta-RL consists of an inner-loop, where the agent interacts with the environment $\mathcal{M} \sim p(\mathcal{M})$, and an outer-loop for transferring knowledge between tasks. Typically, $\mathcal{M}$ is an MDP that doesn't require memory, serving only the outer-loop, which is what "memory" refers to in Meta-RL studies.

The tasks in which the agent makes decisions based on interaction histories in the inner-loop are not named separately, since the classification of Meta-RL task types (multi-task, multi-task 0-shot, and single-task) is based solely on outer-loop parameters (n_{envs} and n_{eps}) and does not consider inner-loop task types. However, we can classify the agent's memory for these tasks as declarative short-term or long-term memory (see Figure 3).

We introduce an additional decoupling of Meta-RL task types into green (with POMDP inner-loop tasks) and blue (with MDP inner-loop tasks). In the green case, the agent's

memory is required for both skill transfer in the outer-loop and decision-making based on interaction histories in the inner-loop, and therefore within the inner-loop can be considered as a Memory DM. In the blue case, memory is needed only for skill transfer. While this paper focuses on Memory DM tasks, the terminology allows for further classification of various Meta-RL tasks, with POMDP sub-classes highlighted in green. The proposed classification of tasks requiring agent memory is presented in Table 1.

Table 1: Classification of tasks requiring agent memory based on our definitions: green indicates tasks described by the proposed definitions of LTM and STM, while blue indicates those that are not. Meta-RL tasks with a POMDP inner-loop are marked green as they can be classified as Memory DM tasks. POMDP† indicates a Memory DM task considered as an inner-loop task without an outer-loop.

Envs. num.	**Runs num.**	**POMDP**	**Inner-loop task**	**Memory**	**Tasks that require agent memory**	
					Memory DM	
					Long-term memory ($\xi > K$)	Short-term memory ($\xi \leq K$)
$n_{envs} = 1$	$n_{eps} = 1$	Memory DM	POMDP†	Declarative	Long-term memory task	Short-term memory task
					Meta-RL: Outer-loop and inner-loop memory	
					Long-term memory ($\xi > K$)	Short-term memory ($\xi \leq K$)
$n_{envs} = 1$	$n_{eps} > 1$	Meta-RL	POMDP	Procedural	Single-task Meta-RL	Single-task Meta-RL
$n_{envs} > 1$	$n_{eps} = 1$	Meta-RL	POMDP	Procedural	Multi-task 0-shot Meta-RL	Multi-task 0-shot Meta-RL
$n_{envs} > 1$	$n_{eps} > 1$	Meta-RL	POMDP	Procedural	Multi-task Meta-RL	Multi-task Meta-RL
					Meta-RL: Outer-loop memory only	
					No memory ($\xi = 1$)	No memory ($\xi = 1$)
$n_{envs} = 1$	$n_{eps} > 1$	Meta-RL	MDP	Procedural	Single-task Meta-RL	Single-task Meta-RL
$n_{envs} > 1$	$n_{eps} = 1$	Meta-RL	MDP	Procedural	Multi-task 0-shot Meta-RL	Multi-task 0-shot Meta-RL
$n_{envs} > 1$	$n_{eps} > 1$	Meta-RL	MDP	Procedural	Multi-task Meta-RL	Multi-task Meta-RL

5.1 MEMORY-INTENSIVE ENVIRONMENTS

To effectively test a Memory DM agent's use of short-term and long-term memory, it is crucial to design appropriate experiments. Not all environments are suitable for assessing agent memory; for example, omnipresent Atari games (Bellemare et al., 2013) with frame stacking or MuJoCo control tasks (Fu et al., 2021) may yield unrepresentative results. To facilitate the evaluation of agent memory capabilities, we formalize the definition of memory-intensive environments:

Definition 5 (**Memory-intensive environments**). Let $\mathcal{M}_P$ be POMDP and $\Xi = \{\xi_n\} = \{(t_r - t_e - \Delta t + 1)_n\}_n$ – set of correlation horizons ξ between for all event-recall pairs. Then $\tilde{\mathcal{M}}_P$ – *memory-intensive* environment $\iff \min_n \Xi > 1$.

Corollary: $\max_n \Xi = 1 \iff \mathcal{M}$ – MDP.

Using the definitions of memory-intensive environments (Definition 5) and agent memory types (Definition 4), we can configure experiments to test short-term and long-term memory in the Memory DM framework. Notably, the same memory-intensive environment can validate both types of memory, as outlined in Theorem 1:

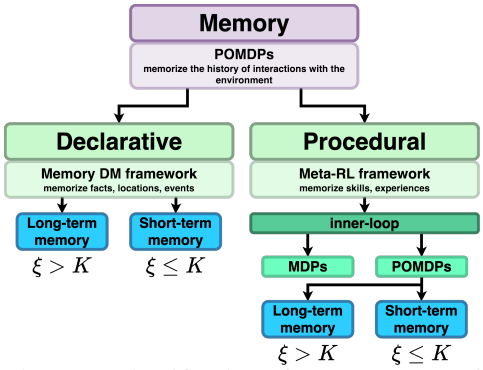


Figure 3: Classification of memory types of RL agents. While the Memory DM framework contrasts with Meta-RL, its formalism can also describe inner-loop tasks when they are POMDPs.

Theorem 1 (**On the context memory border**). *Let $\tilde{\mathcal{M}}_P$ be a memory-intensive environment and K be an agents context length. Then there exists context memory border $\overline{K} \geq 1$ such that if $K \leq \overline{K}$ then the environment $\tilde{\mathcal{M}}_P$ is used to validate exclusively long-term memory in Memory DM framework:*

$$\exists\, \overline{K} \geq 1 : \forall\, K \in [1, \overline{K}] : K < \min_n \Xi \tag{3}$$

Proof. Let $\overline{K} = \min \Xi - 1$. Then $\forall\, K \leq \overline{K}$ is guaranteed that no correlation horizon ξ is in the agent history $h_{t-K+1:t}$, hence the context length $K \leq \min \Xi - 1$ generates the long-term memory problem exclusively. Since context length cannot be negative or zero, it turns out that $1 \leq K \leq \overline{K} = \min \Xi - 1$, which was required to prove. ■

According to Theorem 1, in a memory-intensive environment $\tilde{\mathcal{M}}_P$, the value of the context memory border $\overline{K}$ can be found as

$$\overline{K} = \min \Xi - 1 = \min_n \left\{ (t_r - t_e - \Delta t + 1)_n \right\}_n - 1 \tag{4}$$

Using Theorem 1, we can establish the necessary conditions for validating short-term memory:

1. **Weak condition to validate short-term memory**: if $\overline{K} < K < \max \Xi$, then the memory-intensive environment $\tilde{M}_P$ is used to validate both short-term and long-term memory.
2. **Strong condition to validate short-term memory**: if $\max \Xi < K$, then the memory-intensive environment $\tilde{M}_P$ is used to validate exclusively short-term memory.

According to Theorem 1, if $K \in [1, \overline{K}]$, none of the correlation horizons ξ will be in the agent's context, validating only long-term memory. When $\overline{K} < K < \max \Xi \leq T - 1$, long-term memory can still be tested, but some correlation horizons ξ will fall within the agent's context and won't be used for long-term memory validation. In such a case it is not possible to estimate long-term memory explicitly. When $K \geq \max \Xi$, all correlation horizons ξ are within the agent's context, validating only short-term memory. Summarizing the obtained results, the final division of the required agent context lengths for short-term memory and long-term memory validation is as follows:

Agent context length intervals for separate LTM and STM validation

1. $K \in [1, \overline{K}] \Rightarrow$ validating long-term memory only.
2. $K \in (\overline{K}, \max \Xi) \Rightarrow$ validating both short-term memory and long-term memory.
3. $K \in [\max \Xi, \infty) \Rightarrow$ validating short-term memory only.

5.2 Long-term memory in Memory DM

As defined in Definition 4, Memory DM tasks with short-term memory occur when event-recall pairs in the memory-intensive environment $\tilde{\mathcal{M}}_P$ are within the agent's context ($\xi \leq K$). Here, memory involves the agent's ability to connect information within a context, regardless of how large K is. Examples include works like Esslinger et al. (2022); Ni et al. (2023); Grigsby et al. (2024). Validating short-term memory is straightforward by simply setting a sufficiently large context length K. However, validating long-term memory capabilities is more complex and of greater interest.

Memory DM tasks requiring long-term memory occur when event-recall pairs in the memory-intensive environment $\tilde{\mathcal{M}}_P$ are outside the agent's context ($\xi > K$). In this case, memory involves the agent's ability to connect information beyond its context, necessitating memory mechanisms (Definition 6) that can manage interaction histories h longer than the agent's base model can handle.

Definition 6 (**Memory mechanisms**). Let the agent process histories $h_{t-K+1:t}$ of length K at the current time t, where $K \in \mathbb{N}$ is agents context length. Then, a **memory mechanism** $\mu(K) : \mathbb{N} \to \mathbb{N}$ is defined as a function that, for a fixed K, allows the agent to process sequences of length $K_{eff} \geq K$, i.e., to establish global correlations out of context, where K_{eff} is the *effective context*.

$$\mu(K) = K_{eff} \geq K \tag{5}$$

Memory mechanisms are essential for addressing long-term memory challenges (processing out-of-context information) in the Memory DM framework.

Example of memory mechanism. Consider an agent based on an RNN architecture that can process $K = 1$ triplets of tokens (observations, actions, and rewards) at all times t. By using memory mechanisms $\mu(K)$ (e.g., as in Hausknecht & Stone (2015)), the agent can increase the number of tokens processed in a single step without expanding the context size of its RNN architecture. Therefore, if initially in a memory-intensive environment $\tilde{\mathcal{M}}_P : \xi > K = 1$, it can now be represented as $\tilde{\mathcal{M}}_P : \xi \leq K_{eff} = \mu(K)$. Here, the memory mechanism $\mu(K)$ refers to the RNNs recurrent updates to its hidden state.

Thus, validating an agent's ability to solve long-term memory problems in the Memory DM framework reduces to validating the agent's memory mechanisms $\mu(K)$. **To design correct experiments in such a case, the following condition must be met:**

$$\tilde{\mathcal{M}}_P : K \leq \overline{K} < \xi \leq K_{eff} = \mu(K) \tag{6}$$

According to our definitions, agents with memory mechanisms within the Memory DM framework that can solve long-term memory tasks can also handle short-term memory tasks, but not vice versa. The algorithm for setting up experiments to test an agent's short-term or long-term memory is outlined in Algorithm 1.

Algorithm 1: Algorithm for setting up an experiment to test long-term and short-term memory in Memory DM framework.

Require: $\tilde{\mathcal{M}}_P$ – memory-intensive environment; $\mu(K)$ – memory mechanism.

1. Estimate the number of n event-recall pairs in the environment (Definition 5).

1. $n = 0 \rightarrow$ Environment is not suitable for testing long-term and short-term memory.
2. $n \geq 1 \rightarrow$ Environment is suitable for testing long-term and short-term memory.

2. Estimate context memory border $\overline{K}$ (Equation 4).

1. $\forall$ event-recall pair $(\beta(\alpha), \alpha)_i$ find corresponding $\xi_i, i \in [1..n]$.
2. Determine $\overline{K}$ as $\overline{K} = \min \Xi - 1 = \min_n \{\xi_n\}_n - 1 = \min_n \left\{(t_r - t_e - \Delta t + 1)_n\right\}_n - 1$

3. Conduct an appropriate experiment (Definition 4).

1. To test short-term memory set $K > \overline{K}$.
2. To test long-term memory set $K \leq \overline{K} \leq K_{eff} = \mu(K)$.

4. Analyze the results.

Thus, memory an intrinsic mechanism of the memory-enhanced agent, as it represents the agent's capacity to retain, process, and recall information over time. However, the necessity for memory arises from the requirements of the environment. Thus, memory is considered an intrinsic attribute of an agent, making the classification of memory types inherently tied to the agent itself. However, accurately assessing these memory types requires carefully designed experiments in memory-intensive environments. Using the Algorithm 1, these environments must be configured to challenge the agent's memory mechanisms appropriately, ensuring a clear distinction between short-term and long-term memory capabilities.

5.3 Examples of setting up an experiment to test memory in Memory DM framework

Passive T-Maze. Consider the Passive T-Maze environment (Ni et al., 2023), where the agent starts at the beginning of a T-shaped corridor and observes a clue that is only available at that location. To complete the episode, the agent must walk straight to the junction and turn based on the initial clue. This environment is defined by the corridor length L, with episode duration $T = L + 1$. We will analyze this environment using the Algorithm 1:

1. There is only one event-recall pair in the environment (observing a clue – turning at the junction), so $n = 1$, making it suitable for testing both long-term and short-term memory.
2. The duration of this event is $\Delta t = 0$ (the clue available only at one timestep), and the correlation horizon $\xi = T - 1 - 0 + 1 = T$ (clue at $t = t_e = 0$ and decision-making at $t = t_r = T - 1$). Thus, $\overline{K} = \min_n \{\xi_n\}_n - 1 = T - 1$.
3. By varying the environment parameter $T = L + 1$ or the agent's context size K, we can assess the agent's long-term or short-term memory. For instance, if T is fixed, setting $K > \overline{K} = T - 1$ tests short-term memory. To evaluate long-term memory, we must use memory mechanisms $\mu(K)$ and set context length $K \leq \overline{K} = T - 1 \leq K_{eff} = \mu(K)$.

Theoretically, this estimate $K = \overline{K}$ is sufficient to test the long-term memory of an agent, but in practice it is better to choose a value K closer to the left boundary of the interval $[1, \overline{K}]$, as this allows us to track the effect of the memory mechanism $\mu(K)$ more explicitly.

6 EXPERIMENTS

To illustrate the importance of following a consistent methodology (Algorithm 1) when evaluating an agent's long-term and short-term memory capabilities, as well as to highlight the ambiguity in results that can arise from experimental misconfigurations, we conducted a series of experiments with memory-enhanced agents in memory-intensive environments within the Memory DM framework.

For our experiments, we chose two memory-intensive environments: Passive-T-Maze (Ni et al., 2023) and Minigrid-Memory (Chevalier-Boisvert et al., 2023) (see Appendix, Figure 6). In the Passive-T-Maze, the agent starts at the beginning of a T-shaped maze and observes a clue, which it must use to make a turn at a junction at the end of the maze. The Minigrid-Memory environment presents a similar challenge to the Passive-T-Maze; however, the agent must first reach a room containing a clue before walking down a corridor and making a turn. A detailed description of these environments can be found in Appendix, subsection E.1.

As memory-enhanced baselines, we chose Deep Transformer Q-Networks (DTQN) (Esslinger et al., 2022), DQN with GPT-2 (DQN-GPT-2) (Ni et al., 2023), and Soft Actor-Critic with GPT-2 (SAC-GPT-2) (Ni et al., 2023).

6.1 IMPACT OF EXPERIMENT CONFIGURATION ON MEMORY TYPE TESTED

In subsection 5.1, we identified intervals of agent context length to separate the impact of long-term memory (LTM) and short-term memory (STM). However, the transition between LTM and STM creates an intermediate range where their contributions cannot be clearly distinguished, as some correlation horizons fall inside the agent's context and others do not.

Without standardized definitions or validation methods for LTM and STM, experiments often occur in this transitional interval, making it impossible to assess LTM memory. This ambiguity can lead to misinterpretations of the agent's LTM capabilities, as demonstrated below.

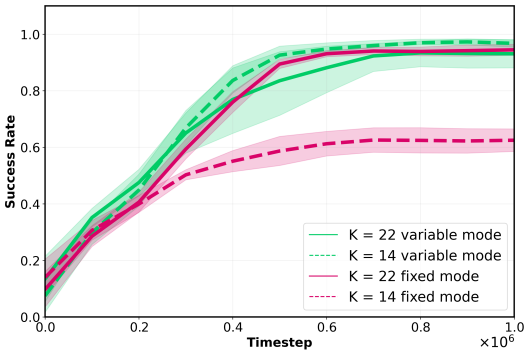


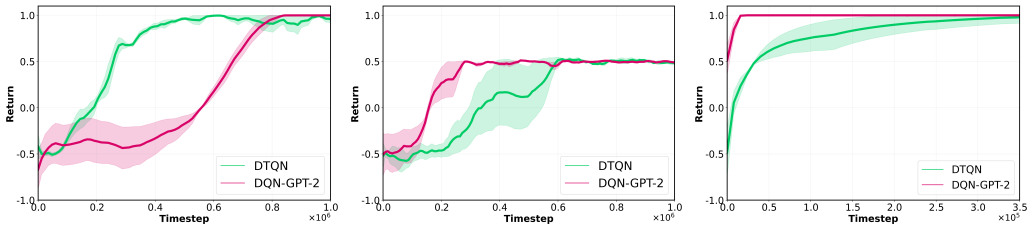
Figure 4: Success Rates for SAC-GPT-2 agent with LTM and STM for the Minigrid-Memory environment with map size $L = 21$.

To illustrate this, we conducted experiments with the transformer-based agent SAC-GPT-2 in the MiniGrid-Memory environment, setting the map size to $L = 21$. Two experimental configurations were used: fixed-length corridors with $\xi = L + 1$ (fixed mode) and variable-length corridors with $\xi \in [7, L + 1]$ (variable mode). If the methodology proposed in Algorithm 1 for testing LTM and STM within the Memory DM framework is not followed, the agent's context length K might be set arbitrarily as $K = 14$ (representing LTM, since $K < L$) or $K = 22$ (representing STM, since $K > L$).

The results of this experiment are shown in Figure 4. The solid line represents STM ($K = 22$), the dashed line represents LTM ($K = 14$), while green indicates variable mode and red indicates fixed mode. In variable mode (green), the agent achieves a success rate (SR) almost 1.0 for both LTM and STM validation experiments. This might incorrectly suggest that the agent possesses both memory types. Conversely, in fixed mode (red), the results reveal a discrepancy: the agent demonstrates STM memory but fails to exhibit LTM memory.

This discrepancy arises because SAC-GPT-2 lacks memory mechanisms to solve LTM problems; it can only leverage information within its context K. The confusion occurs due to a naive experimental setup, where K was chosen relative to L based solely on environmental documentation, without consideration for the interaction of LTM and STM. In variable mode, the agent's performance reflects a mix of LTM and STM capabilities, making it impossible to isolate LTM memory explicitly. In contrast, the fixed mode, tested according to the methodology outlined in Algorithm 1, clearly identifies STM memory while confirming the absence of LTM memory.

In this section, we have demonstrated that a naive approach to testing an agent's memory can result in misinterpreting its true capabilities. In contrast, our proposed methodology enables the design of



$K = 15,\ \xi = 15$ $K = 5,\ \xi = 15$ $K = 5,\ \xi = 5$

Figure 5: Results for the DQN-GPT-2 and DTQN agents in the Passive-T-Maze. The STM $\leftrightarrows$ LTM transitions reflect the relative nature of the settings to test memory, depending on both agent and environment parameters: STM with $K = 15$, to LTM with $K = 5$, and back to STM with $K = 5$.

experiments that explicitly distinguish and accurately evaluate the agent's long-term memory and short-term memory.

6.2 The relative nature of an agent's memory

According to the Algorithm 1, the experimental setup for testing agent memory types (LTM and STM) relies on two parameters: the agent's context length K and the context memory border $\overline{K}$, which depends on the environment properties, ξ. Verifying an agent's LTM or STM requires adjusting K or ξ while keeping the other fixed. This section explains how these parameters interact in memory testing experiments.

We evaluate two memory-enhanced agents, DTQN and DQN-GPT-2, in the Passive T-Maze environment by varying K and ξ. The results are shown in Figure 5.

First, we test STM by setting $K = \xi = 15$. In this configuration, all relevant information stays within the agent's context. As shown in Figure 5 (left), both agents achieve a return of 1.0, confirming STM capabilities. To test LTM, we use $\xi = 15$ and adjust the setup so that key event-recall pairs fall outside the agent's context. By reducing K from 15 to 5, as shown in Figure 5 (center), the return of both agents drops to 0.5, indicating that they cannot recall the cue information, which confirms that LTM is not LTM. Next, we further assess STM by reducing ξ. With $K = 5$ and ξ reduced from 15 to 5, as shown in Figure 5 (right), the agent's return returns to 1.0. This shows that agents can effectively use memory when all relevant information is within their context.

In summary, verifying LTM and STM can be done by adjusting K or ξ while keeping the other fixed. The Passive T-Maze is an effective testbed due to its parameterizable corridor length L, which relates to ξ as $\xi = L + 1$. However, in many environments where ξ is fixed, varying K remains a viable approach for memory evaluation.

7 Conclusion

In this study, we formalize memory types in RL, distinguishing long-term memory (LTM) from short-term memory (STM), and declarative from procedural memory, drawing inspiration from neuroscience. We also separate POMDPs into two classes: Memory Decision-Making (Memory DM) and Meta Reinforcement Learning (Meta-RL).

The formalization, along with the methodology for validating LTM and STM in the Memory DM framework, provides a clear structure for distinguishing between different types of agent memory. This enables fair comparisons of agents with similar memory mechanisms and highlights limitations in memory architecture, facilitating precise evaluations and improvements.

Additionally, we demonstrate the potential pitfalls of neglecting this methodology. Misconfigured experiments can lead to misleading conclusions about an agent's memory capabilities, blurring the lines between LTM and STM. By following our approach, researchers can achieve more reliable assessments and make informed comparisons between memory-enhanced agents.

This work provides a significant step toward a unified understanding of agent memory in RL. Our definitions and methodology offer practical tools for rigorously testing agent memory, ensuring consistent experimental design. By addressing common inconsistencies, our approach guarantees reliable results and meaningful comparisons, advancing research in RL.

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A Appendix – Glossary

In this section, we provide a comprehensive glossary of key terms and concepts used throughout this paper. The definitions are intended to clarify the terminology proposed in our research and to ensure that readers have a clear understanding of the main elements underpinning our work.

1. $\mathcal{M}$ – MDP environment
2. $\mathcal{M}_P$ – POMDP environment
3. $\tilde{\mathcal{M}}_P$ – memory-intensive environment
4. $h_{0:t-1} = \{(o_i, a_i, r_i)\}_{i=0}^{t-1}$ – agent history of interactions with environment
5. K – agent base model context length
6. $\overline{K}$ – context memory border of the agent, such that $K \in [1, \overline{K}] \Leftrightarrow$ strictly LTM problem
7. $\mu(K)$ – memory mechanism that increases number of steps available to the agent to process
8. $K_{eff} = \mu(K)$ – the agent effective context after applying the memory mechanism
9. $\alpha_{t_e}^{\Delta t} = \{(o_i, a_i, r_i)\}_{i=t_e}^{t_e+\Delta t}$ – an event starting at time t_e and lasting Δt, which the agent should recall when making a decision in the future
10. $\beta_{t_r} = \beta_{t_r}(\alpha_{t_e}^{\Delta t}) = a_t \mid (o_t, \alpha_{t_e}^{\Delta t})$ – the moment of decision making at time t_r according to the event $\alpha_{t_e}^{\Delta t}$
11. $\xi = t_r - t_a - \Delta t + 1$ – an event's correlation horizon

B Appendix – Additional notes on the motivation for the article

B.1 Why use definitions from neuroscience?

Definitions from neuroscience and cognitive science, such as short-term and long-term memory, as well as declarative and procedural memory, are already well-established in the RL community, but do not have common meanings and are interpreted in different ways. We strictly formalize these definitions to avoid possible confusion that may arise when introducing new concepts and redefine them with clear, quantitative meanings to specify the type of agent memory, since the performance of many algorithms depends on their type of memory.

In focusing exclusively on memory within RL, we do not attempt to exhaustively replicate the full spectrum of human memory. Instead, our goal is to leverage the intuitive understanding of neuroscience concepts already familiar to RL researchers. This approach avoids the unnecessary introduction of new terminology into the already complex Memory RL domain. By refining and aligning existing definitions, we create a robust framework that facilitates clear communication, rigorous evaluation, and practical application in RL research.

B.2 On practical applications of our framework

The primary goal of our framework is to address practical challenges in RL by providing a robust classification of memory types based on temporal dependencies and the nature of memorized information. This classification is essential for standardizing memory testing and ensuring that RL agents are evaluated under conditions that accurately reflect their capabilities.

In RL, memory is interpreted in various ways, such as transformers with large context windows, recurrent networks, or models capable of skill transfer across tasks. However, these approaches often vary fundamentally in design, making comparisons unreliable and leading to inconsistencies in testing. Our framework resolves this by providing a clear structure to evaluate memory mechanisms under uniform and practical conditions.

The proposed definitions of declarative and procedural memory use two straightforward numerical parameters: the number of environments (n_{envs}) and episodes (n_{eps}). These parameters allow researchers to reliably determine the type of memory required for a task. This simplicity and alignment with numerical parameters make the framework practical and widely applicable across diverse RL problems.

Moreover, the division of declarative memory into long-term and short-term memory, as well as the need to use a balance between the agent's context length K and the correlation horizons of the environment ξ when conducting the experiment, allows us to unambiguously determine which type of memory is present in the agent. This clarity ensures fair comparisons between agents with similar memory mechanisms and highlights specific limitations in an agent's design. By aligning memory definitions with practical testing requirements, the framework provides actionable insights to guide the development of memory-enhanced RL agents.

C Appendix – Memory Mechanisms

In RL, memory has several meanings, each of which is related to a specific class of different tasks. To solve these tasks, the authors use various memory mechanisms. The most prevalent approach to incorporating memory into an agent is through the use of Recurrent Neural Networks (RNNs) (Rumelhart et al., 1986), which are capable of handling sequential dependencies by maintaining a hidden state that captures information about previous time steps (Wierstra et al., 2010; Hausknecht & Stone, 2015; Sorokin et al., 2015; Duan et al., 2016; Song et al., 2018; Zintgraf et al., 2020). Another popular way to implement memory is to use Transformers (Vaswani et al., 2017), which use self-attention mechanisms to capture dependencies inside the context window (Parisotto et al., 2020; Lampinen et al., 2021; Esslinger et al., 2022; Melo, 2022; Team et al., 2023; Pramanik et al., 2023; Robine et al., 2023; Ni et al., 2023; Grigsby et al., 2024; Shala et al., 2024). State-space models (SSMs) (Gu et al., 2021; Smith et al., 2023; Gu & Dao, 2023) combine the strengths of RNNs and Transformers and can also serve to implement memory through preservation of system state (Hafner et al., 2019; Lu et al., 2023; Becker et al., 2024; Samsami et al., 2024). Temporal convolutions may be regarded as an effective memory mechanism, whereby information is stored implicitly through the application of learnable filters across the time axis (YuXuan Liu & Hsieh, 2016; Mishra et al., 2018). A world model (Ha & Schmidhuber, 2018) which builds an internal environment representation can also be considered as a form of memory. One method for organizing this internal representation is through the use of a graph, where nodes represent observations within the environment and edges represent actions (Morad et al., 2021; Zhu et al., 2023; Kang et al., 2024b).

A distinct natural realization of memory is the utilization of an external memory buffer, which enables the agent to retrieve pertinent information. This approach can be classified into two categories: read-only (writeless) (Oh et al., 2016; Lampinen et al., 2021; Goyal et al., 2022; Cherepanov et al., 2024) and read/write access (Graves et al., 2016; Zaremba & Sutskever, 2016; Parisotto & Salakhutdinov, 2017).

Memory can also be implemented without architectural mechanisms, relying instead on agent policy. For instance, in the work of Deverett et al. (2019), the agent learns to encode temporal intervals by generating specific action patterns. This approach allows the agent to implicitly represent timing information within its behavior, showcasing that memory can emerge as a result of policy adaptations rather than being explicitly embedded in the underlying neural architecture.

Using these memory mechanisms, both decision-making tasks based on information from the past within a single episode and tasks of fast adaptation to new tasks are solved. However, even in works using the same underlying base architectures to solve the same class of problems, the concepts of memory may differ.

D Appendix – Meta Reinforcement Learning

In this section, we explore the concept of Meta-Reinforcement Learning (Meta-RL), a specialized domain within POMDPs that focuses on equipping agents with the ability to learn from their past experiences across multiple tasks. This capability is particularly crucial in dynamic environments where agents must adapt quickly to new challenges. By recognizing and memorizing common patterns

and structures from previous interactions, agents can enhance their efficiency and effectiveness when facing unseen tasks.

Meta-RL is characterized by the principle of "*learning to learn*", where agents are trained not only to excel at specific tasks but also to generalize their knowledge and rapidly adjust to new tasks with minimal additional training. This adaptability is achieved through a structured approach that involves mapping data collected from various tasks to policies that guide the agent's behavior.

Meta-RL algorithm is a function f_θ parameterized with *meta-parameters* that maps the data $\mathcal{D}$, obtained during the process of training of RL agent in MDPs (tasks) $\mathcal{M}_i \sim p(\mathcal{M})$, to a policy $\pi_\phi : \phi = f_\theta(\mathcal{D})$. The process of learning the function f is typically referred to as the *outer-loop*, while the resulting function f is called the *inner-loop*. In this context, the parameters θ are associated with the outer-loop, while the parameters ϕ are associated with the inner-loop. Meta-training proceeds by sampling a task from the task distribution, running the inner-loop on it, and optimizing the inner-loop to improve the policies it produces. The interaction of the inner-loop with the task, during which the adaptation happens, is called a *lifetime* or a *trial*. In Meta-RL, it is common for $\mathcal{S}$ and $\mathcal{A}$ to be shared between all of the tasks and the tasks to only differ in the reward $\mathcal{R}(s, a)$ function, the dynamics $\mathcal{P}(s^{'} \mid s, a)$, and initial state distributions $P_0(s_0)$ (Beck et al., 2024). The formal definition of Meta-RL framework is presented in Definition 7.

Definition 7 (**Meta-RL**). **Meta-RL** – is a class of POMDPs where the agent learns to learn from its past experiences across multiple tasks and memorize the common patterns and structures to facilitate efficient adaptation to new tasks. Let $\mathcal{D} = \{\tau_j^{\mathcal{M}_i}\}_{j=0}^{H-1}$ is all of the data of H episodes of length T collected in the MDP $\mathcal{M}_i \sim p(\mathcal{M})$. A Meta-RL algorithm is a function f_θ that maps the data $\mathcal{D}$ to a policy π_ϕ, where $\phi = f_\theta(\mathcal{D})$. The objective to determine an optimal f_θ: $J^\theta = \mathbb{E}_{\mathcal{M}_i \sim p(\mathcal{M})}\left[\mathbb{E}_{\mathcal{D}}\left[\sum_{\tau \in \mathcal{D}_{I:H}} G_i(\tau) \middle| f_\theta, \mathcal{M}_i\right]\right]$, where $G_i(\tau)$ – discounted return in the MDP $\mathcal{M}_i$, I – index of the first episode during the trial in which return counts towards the objective (Beck et al., 2024).

E Appendix – Experiment Details

E.1 Appendix – Environments description

This section provides an extended description of the environments used in this work.

Passive-T-Maze (Ni et al., 2023). In this T-shaped maze environment, the agent's goal is to move from the starting point to the junction and make the correct turn based on an initial signal. The agent can select from four possible actions: $a \in left, up, right, down$. The signal, denoted by the variable $clue$, is provided only at the beginning of the trajectory and indicates whether the agent should turn up ($clue = 1$) or down ($clue = -1$). The episode duration is constrained to $T = L+1$, where L is the length of the corridor leading to the junction, which adds complexity to the task. To facilitate navigation, a binary variable called $flag$ is included in the observation vector. This variable equals 1 one step before reaching the junction and 0 at all other times, indicating the agent's proximity to the junction. Additionally, a noise channel introduces random integer values from the set $-1, 0, +1$ into the observation vector, further complicating the task. The observation vector is defined as $o = [y, clue, flag, noise]$, where y represents the vertical coordinate.

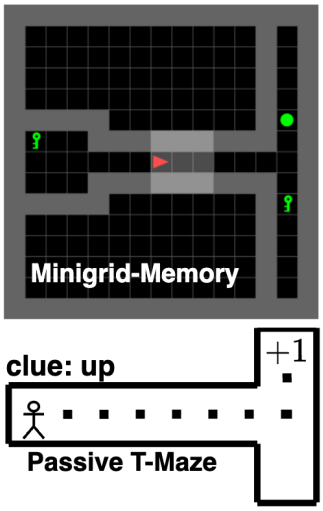

Figure 6: Memory-intensive environments for testing STM and LTM in Memory DM.

The agent receives a reward only at the end of the episode, which depends on whether it makes a correct turn at the junction. A correct turn yields a reward of 1, while an incorrect turn results in a reward of 0. This configuration differs from the conventional Passive T-Maze environment (Ni et al., 2023) by featuring distinct

observations and reward structures, thereby presenting a more intricate set of conditions for the agent to navigate and learn within a defined time constraint. To transition from a sparse reward function to a dense reward function, the environment is parameterized using a penalty defined as $penalty = -\frac{1}{T-1}$, which imposes a penalty on the agent for each step taken within the environment. Thus, this environment has a 1D vector space of observations, a discrete action space, and sparse and dense configurations of the reward function.

Minigrid-Memory (Chevalier-Boisvert et al., 2023). Minigrid-Memory is a two-dimensional grid-based environment specifically crafted to evaluate an agent's long-term memory and credit assignment capabilities. The layout consists of a T-shaped maze featuring a small room at the corridor's outset, which contains an object. The agent is instantiated at a random position within the corridor. Its objective is to navigate to the chamber, observe and memorize the object, then proceed to the junction at the maze's terminus and turn towards the direction where the object, identical to that in the initial chamber, is situated. A reward function defined as $r = 1 - 0.9 \times \frac{t}{T}$ is awarded upon successful completion, while failure results in a reward of zero. The episode concludes when the agent either makes a turn at a junction or exhausts a predefined time limit of 95 steps. To implement partial observability, observational constraints are imposed on the agent, limiting its view to a 3×3 frame size. Thus, this environment has a 2D space of image observations, a discrete action space, and sparse reward function.

E.2 Experimental Protocol

For each experiment, we conducted three runs of the agents with different initializations and performed validation during training using 100 random seeds ranging from 0 to 99. The results are presented as the mean success rate (or reward) ± the standard error of the mean (SEM).

Table 2: Hyperparameters used in the Minigrid-Memory and Passive T-Maze experiments.

(a) SAC-GPT-2

Hyperparameter	**Value**
Number of layers	2
Number of attention heads	2
Hidden dimension	256
Batch size	64
Optimizer	Adam
Learning rate	3e-4
Dropout	0.1
Replay buffer size	1e6
Discount (γ)	0.99
Entropy temperature	0.1

(b) DQN-GPT-2

Hyperparameter	**Value**
Number of layers	2
Number of attention heads	2
Hidden dimension	256
Batch size	64
Optimizer	Adam
Learning rate	3e-4
Dropout	0.1
Replay buffer size	1e6
Discount (γ)	0.99

(c) DTQN

Hyperparameter	**Value**
Number of layers	4
Number of attention heads	8
Hidden dimension	128
Batch size	32
Optimizer	Adam
Learning rate	3e-4
Dropout	0.1
Replay buffer size	5e5
Discount (γ)	0.99